Sales Data Quality and Fitness Assessment

An independent review of **sales_data_large.csv** carried out before any dashboard was built, to establish which business questions this dataset can and cannot answer.

> **VERDICT**
> This dataset is technically flawless and analytically empty. It passed every standard data-cleaning check with zero defects, then failed every test that decides whether a number is worth acting on.

THE FILE AT A GLANCE

Measure	Value	Measure	Value
File assessed	sales_data_large.csv	Cleaning defects found	0 of 8 checks failed
Rows and columns	1,000 x 9	Columns that fail key integrity	**3 of 9**
Period covered	1 Jan to 31 Dec 2024	Significance tests run	9 marginal, 7 deep angles
Days carrying a sale	345 of 366	Tests returning a signal	**0**
Total revenue, units	2,561,723 from 9,746	Revenue range	101 to 4,995
Transactions	1,000, averaging 2,562	Decisions the file supports	**None beyond totals**

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Prepared from the delivered file only. No figure here is estimated or carried over from a previous engagement; every one is reproducible from the scripts listed on page 10.

Executive summary

> **KEY FINDING**
>
> This dataset is technically flawless and analytically empty. It passed every standard data-cleaning check with zero defects. It then failed the tests that matter: three of its identifier and measure columns do not hold together, and every remaining dimension is statistically indistinguishable from random noise. Nine significance tests and seven further analytical angles were run in all, from interaction effects and bootstrap confidence intervals to Benford's law and a power calculation, and none of them found a decision the data could support. **No sales dashboard built on this file can support a 'where should we focus' decision, because the file contains nothing to focus on.**

We assessed the file in two passes. The first was the conventional cleaning checklist: types, headers, missing values, duplicates, outliers, low-variance columns and date continuity. The file passed all of it cleanly, which is why the problems described here would survive an ordinary quality review unnoticed.

The second pass tested whether the columns mean what their names imply, and whether the differences between categories are larger than chance would produce. Both tests failed. The findings below are ordered by how much they constrain what can be delivered.

What this means for the questions usually put to sales data

Question	Answerable?	The measured reason
Which city should we prioritise?	**No**	The best-to-worst gap is 1.43x, reproduced by chance in 35% of 20,000 simulations. Every city's 95% interval overlaps every other city's.
Which category should we push?	**No**	Chi-square 2.93 on 4 degrees of freedom, p = 0.57. The five categories run from 18.1% to 21.1% against an even split of 20.0%.
Who are our best customers?	**No**	258 of the 631 CustomerIDs carry conflicting age, gender or city, so no customer can be followed from one row to the next.
Which products earn most?	**No**	All 100 ProductIDs appear under more than one category, so the identifier does not name a product.
What is our margin?	**No**	There is no cost column. This is a structural absence, not a broken column, and no proxy should be substituted for it.
What will next quarter look like?	**No**	The trend is +0.46% of an average month and the lag-1 autocorrelation is -0.468, so a good month does not predict the next.
How much did we sell in 2024?	**Yes**	2,561,723 across 1,000 transactions and 9,746 units. The totals are sound. It is the comparisons between them that are not.

The pattern in that table is itself the finding: everything the file can answer is a total, and everything it cannot answer is a comparison. A dashboard is mostly comparisons.

1. How the assessment was run

The review followed a fixed sequence, and each step was allowed to stop the next. Nothing was reported as a finding until a test had been run against it. The sequence matters because this file fails at steps 2 and 3, both of which sit after the checklist that most reviews stop at.

Step	What it tests	Outcome on this file
0. Classify	Shape and grain of the delivered data	A single flat transaction table with a date column. 1,000 rows over 345 active days is 2.9 per day, so the natural grain is monthly, not daily.
1. Clean	Mechanical defects: types, blanks, duplicates, outliers	Passed. Zero defects across all eight checks, detailed below.
2. Key integrity	Whether each column means what its name implies	**Failed on three columns. See page 4.**
3. Marginal tests	Whether the observed differences beat chance	**Nine tests, nine null results. See page 5.**
4. Deep battery	Whether any angle holds signal the totals would hide	**Seven independent angles, seven null results. See pages 7 and 8.**
5. Scope	What may honestly be published	Totals and shares only. No ranking, no prioritisation, no forecast. See page 10.

The cleaning checklist, in full

This is the conventional quality review, run first and reported here in full because its result is counter-intuitive. The file is in better mechanical condition than most production exports.

Cleaning check	Result
Row and column count	1,000 rows, 9 columns
Type conformance across all typed columns	0 non-conforming values in SalesDate, QuantitySold, SalesRevenue and CustomerAge
Header integrity	Clean, no title row above the header
Missing values and disguised nulls	0 blank cells, 0 sentinel values
Exact and near-duplicate rows	0 exact, 0 near-duplicates on date plus product plus customer
Outliers and implausible values	None. Revenue 101 to 4,995, quantity 1 to 19, ages 18 to 69, no negatives or zeroes
Constant or near-constant columns	None above the 95% threshold
Date continuity across 2024	345 of 366 days carry a sale, 21 do not, longest consecutive gap 1 day

WHY THE CHECKLIST IS NOT ENOUGH

A clean bill of health from the standard checklist is exactly why the real problems here would survive an ordinary review. Cleanliness is a property of the file. Fitness is a property of the file measured against a question. They are different tests, and this file separates them completely.

2. Finding 1: three columns do not mean what they are named

Column names imply a structure the data does not actually carry. Each issue below is invisible in the header row and in any spot check of the first few records. All three appear only when consistency is tested across the whole file.

Column	What the name implies	What the data actually shows	Consequence
ProductID	A stable product key, 100 products	All 100 IDs appear under more than one product category	No product-level analysis is possible. Use **ProductCategory** instead.
CustomerID	A stable customer key, 631 customers	258 of the 631 IDs carry conflicting age, gender or city	No customer view, no repeat-purchase or churn analysis.
SalesRevenue	Revenue earned on the units sold	Uncorrelated with QuantitySold (r = -0.06)	Unit price, basket value and any per-unit metric are meaningless.

A fourth constraint is structural rather than broken: the file carries no cost column, so profit and margin cannot be derived at all.

The evidence behind each one

Column	Test applied across all 1,000 rows	Measured	A usable key would show
ProductID	IDs appearing under more than one ProductCategory	**100 of 100**	0 of 100
CustomerID	IDs carrying a conflicting age, gender or city	**258 of 631**	0 of 631
SalesRevenue	Pearson correlation with QuantitySold	**r = -0.0599**	Strongly positive
Cost or unit price	Presence of any cost, price or margin column	**Absent**	Present

What this blocks, stated plainly

These are not panels that would come out weaker than hoped. They cannot be built at all from this file, because the column each one would read does not carry the meaning the panel needs.

Blocked analysis	Root cause
Named product performance, product mix, product drill-down	ProductID is not a stable key
Customer profile, segmentation, lifetime value	CustomerID is not a stable key
Repeat purchase, retention, churn	Follows from the CustomerID failure
Average selling price, basket value, unit economics	Revenue is uncorrelated with quantity
Gross profit, margin percentage, contribution	No cost column exists
Discount or promotion effect	No price or discount column exists

The remedy for all six is the same, and it is not analytical: a production export carrying a stable key per product, a stable record per customer, and a cost column. Recommendation 2 on page 10 states the full specification.

3. Finding 2: the data contains no measurable signal

Revenue is spread almost perfectly evenly across every available dimension. We tested whether the small differences that do appear are larger than random variation would produce. Across nine separate tests, they are not. The register below is the complete list. Nothing was run and then left out of it.

#	Test	Statistic	p	Verdict
1	Revenue by product category, 5 values	chi-square 2.93, df 4	0.570	**No real difference**
2	Revenue by city, 10 values	chi-square 6.20, df 9	0.720	**No real difference**
3	Revenue by gender, 2 values	chi-square 0.26, df 1	0.610	**No real difference**
4	Revenue by customer age band, 5 values	chi-square 3.73, df 4	0.444	**No real difference**
5	Revenue by day of week, 7 values	chi-square 5.76, df 6	0.451	**No real difference**
6	Category by city interaction on revenue	permutation, 20,000 shuffles	0.184	**No real difference**
7	Gap between highest and lowest city, 1.43x	permutation, 20,000 shuffles	0.345	**Ordinary noise**
8	Month-on-month trend across 2024	+0.46% of an average month	n/a	**Flat, no growth**
9	Leading digits of the revenue values	chi-square 442, df 8, against Benford	2e-90	**Randomly generated**

Tests 1 to 5 compare the observed split against an even split at the 5% level. Test 9 is the only rejection in the register, and it rejects in the direction least useful to the client: it rejects the hypothesis that these amounts arose naturally. Where a family of related tests was run, the Bonferroni-corrected threshold is 0.05 divided by 7, which is 0.0071. No result comes near either threshold.

One mistake, corrected and kept on the record

> **METHOD NOTE**
>
> An early pass tested the category-by-city interaction by computing a chi-square statistic on **summed revenue** and reading the p-value from a chi-square table. That produced **p = 0.023**, which would have been reported as the one real finding in the file. It is wrong. A chi-square statistic computed on summed money is not distributed as chi-square, so the table does not apply to it. Re-running the same comparison as a 20,000-shuffle permutation test, which needs no distributional assumption, gives **p = 0.184**. The rule this produced, and the reason the mistake is printed rather than quietly fixed: when a statistic is not a standard one, obtain the p-value by permutation rather than by looking it up.

The interaction family was tested on transaction counts as well as on revenue, because a flat set of totals can still hide a combination that behaves differently, for example one category booming in one city only. Five count-based contingency tables were tested: category by city (p = 0.408), category by age band (p = 0.679), category by gender (p = 0.153), city by gender (p = 0.907) and city by age band (p = 0.922). The worst of them is 0.153. No combination carries signal that the marginal totals hide.

4. Where the revenue actually sits

These tables are the evidence behind Finding 2, shown as the business would read them. In each case the right-hand columns give the share a perfectly even split would produce.

By product category

Product category	Revenue	Share	Transactions	Units	Even split
Home Decor	540,975	21.1%	205	1,982	20.0%
Beauty	539,969	21.1%	204	2,051	20.0%
Books	520,795	20.3%	212	2,100	20.0%
Apparel	497,068	19.4%	199	1,878	20.0%
Electronics	462,916	18.1%	180	1,735	20.0%

Spread from highest to lowest: 1.17x. Chi-square 2.93 on 4 degrees of freedom, p = 0.57.

By customer city

City	Revenue	Share	Transactions	Even split	Gap to even
Dallas	302,114	11.8%	118	10.0%	+1.8 points
Chicago	282,553	11.0%	107	10.0%	+1.0
Phoenix	267,683	10.4%	102	10.0%	+0.4
San Jose	265,975	10.4%	101	10.0%	+0.4
Los Angeles	259,752	10.1%	96	10.0%	+0.1
Houston	258,297	10.1%	99	10.0%	+0.1
Philadelphia	249,967	9.8%	96	10.0%	-0.2
San Antonio	232,752	9.1%	98	10.0%	-0.9
San Diego	231,078	9.0%	97	10.0%	-1.0
New York	211,552	8.3%	86	10.0%	-1.7

Spread from highest to lowest: 1.43x. Chi-square 6.20 on 9 degrees of freedom, p = 0.72. The order shown here is correct arithmetic on the delivered file, and it is also not reproducible: page 8 shows what happens to it under resampling.

By customer age band and by gender

Age band	Revenue	Share	Even split	Gender	Revenue	Share	Even split
18-29	565,535	22.1%	20.0%	Male	1,295,751	50.6%	50.0%
30-39	545,454	21.3%	20.0%	Female	1,265,972	49.4%	50.0%
40-49	495,830	19.4%	20.0%				
50-59	487,261	19.0%	20.0%				
60+	467,643	18.3%	20.0%				

Age bands: chi-square 3.73, df 4, p = 0.44 against an even 20.0% each. Gender: chi-square 0.26, df 1, p = 0.61 against an even 50.0% each. CustomerAge holds 52 distinct values and was bucketed into five bands before testing.

5. Concentration, and how far we looked

The concentration check makes the same point in business terms, and it is the single most decision-relevant measure in this report. In real sales data the top tenth of customers, products or territories usually carries 40% to 80% of revenue, and that unevenness is what makes prioritisation possible. Here the top tenth carries almost exactly its even share.

Concentration check	Observed	Even split would be	Real business range
Top 10% of cities	11.8% of revenue	10.0%	40% to 80%
Top 10% of product categories	21.1% of revenue	20.0%	40% to 80%
Half of all revenue comes from	5 of the 10 cities	5 of 10	1 or 2 of 10
Half of all revenue comes from	3 of the 5 categories	2.5 of 5	1 of 5

Gini coefficient, 0 is perfectly even	Value	Reading	Real sales data
By city, 10 values	0.055	Essentially even	0.40 to 0.70
By product category, 5 values	0.031	Essentially even	0.40 to 0.70
Across individual transactions	0.323	Consistent with a flat random draw	Higher and lumpy

Seven further angles

Findings 1 and 2 rest on the marginal totals. Before concluding that the file holds no usable signal, we tested seven further angles, each able to find structure the totals would hide.

Angle	What it would have found	Result
Two-way interactions	A combination that performs differently from what its parts predict	All five pairs were tested. Worst p-value 0.15; a permutation test on the category-by-city revenue matrix returned p = 0.18. **No combination carries signal the marginal totals hide.**
Bootstrap confidence intervals	A city whose true revenue is reliably above the others	Across 4,000 resamples every city's 95% interval overlaps every other city's. Dallas is top in only 46% of them, and every city is seen at every rank from 1 to 10. **The city ranking is not stable.**
Concentration, Gini coefficient	The uneven spread that makes prioritisation possible	See the table above. Concentration is what makes prioritisation possible, and **its absence here is why no city and no category can be singled out.**
Benford's law	The leading-digit signature that naturally occurring amounts carry	Rejected at chi-square 442, df 8, p = 2e-90. The leading digits instead match a flat draw between 101 and 4,995 (p = 0.85). **Forensic evidence of generated values.**
Temporal structure	A weekday, weekend or momentum effect worth scheduling against	No day-of-week effect (p = 0.45); weekend share 30.2% of revenue on 28.6% of days; lag-1 autocorrelation -0.47, so **a good month does not predict the next.**
Power analysis	Whether 1,000 rows could detect a real difference even if one existed	At this sample size only large gaps are detectable, so the silence is partly a sample-size limit. **Full specification on page 9.**
Anomaly scan	The exceptional transactions that usually repay investigation	Zero transactions beyond three standard deviations, **which is itself abnormal for genuine sales data.**

Seven independent angles, seven null results. The consistency matters more than any single test: a real dataset that failed one of these would usually pass another.

6. Ranking stability and the forensic test

Two of the seven angles carry enough weight to be shown in full. The first answers the objection that a p-value is an abstraction: it asks directly whether the city ranking on page 6 would come back the same way if the year had gone slightly differently. The second answers the question of where these numbers came from in the first place.

Bootstrap confidence intervals: 4,000 resamples of the 1,000 rows

City	Observed share	95% confidence interval	Contains the 10.0% even split?	Ranks seen
Dallas	11.8%	9.6% to 14.1%	Yes	1 to 10
Chicago	11.0%	8.9% to 13.3%	Yes	1 to 10
Phoenix	10.4%	8.3% to 12.8%	Yes	1 to 10
San Jose	10.4%	8.2% to 12.6%	Yes	1 to 10
Los Angeles	10.1%	8.0% to 12.5%	Yes	1 to 10
Houston	10.1%	8.0% to 12.3%	Yes	1 to 10
Philadelphia	9.8%	7.7% to 11.9%	Yes	1 to 10
San Antonio	9.1%	7.1% to 11.2%	Yes	1 to 10
San Diego	9.0%	7.0% to 11.1%	Yes	1 to 10
New York	8.3%	6.4% to 10.2%	Yes	1 to 10

WHAT THE INTERVALS SAY

Every interval overlaps every other interval, and every interval contains the 10.0% that a perfectly even split would produce. Dallas, the apparent leader, is top in only 46% of the 4,000 resamples, and each of the ten cities is observed at every rank from first to tenth. **The ranking is an artefact of which 1,000 transactions happened to land in the file.**

Benford's law, applied to the leading digit of every revenue amount

Amounts produced by real economic activity carry a characteristic leading-digit signature: a 1 about 30% of the time, falling away to about 5% for a 9. Amounts drawn from a flat range do not. This is the test that moves the assessment from 'looks synthetic' to 'is demonstrably generated'.

Leading digit	1	2	3	4	5	6	7	8	9
Observed count	236	208	229	233	18	20	15	21	20
Observed share	23.6%	20.8%	22.9%	23.3%	1.8%	2.0%	1.5%	2.1%	2.0%
Benford expects	30.1%	17.6%	12.5%	9.7%	7.9%	6.7%	5.8%	5.1%	4.6%
A flat 101 to 4,995 draw expects	22.5%	22.5%	22.5%	22.4%	2.0%	2.0%	2.0%	2.0%	2.0%

Fit against Benford: chi-square 442.0 on 8 degrees of freedom, p = 2e-90, rejected. Fit against a flat random draw between 101 and 4,995: chi-square 4.1 on 8 degrees of freedom, p = 0.851, consistent. The collapse after digit 4 is the signature of a hard upper bound at 4,995 rather than of a price list.

7. Time, and the export that would work

Revenue by month is the one view that visibly moves, so it is the one most likely to be over-interpreted. The monthly series varies by 16.0% around its mean, which reads as seasonality and is not: the fitted trend is +991 per month, or +0.46% of an average month, and the lag-1 autocorrelation is -0.468, meaning a strong month is followed by a weak one slightly more often than not. That is the signature of noise, not of momentum.

Month	Revenue	Txns	Share	Month	Revenue	Txns	Share
2024-01	200,691	76	7.8%	2024-07	175,609	71	6.9%
2024-02	240,151	88	9.4%	2024-08	212,981	95	8.3%
2024-03	187,792	73	7.3%	2024-09	171,612	64	6.7%
2024-04	210,782	86	8.2%	2024-10	273,842	97	10.7%
2024-05	261,895	105	10.2%	2024-11	189,816	76	7.4%
2024-06	191,113	78	7.5%	2024-12	245,439	91	9.6%

Monthly mean 213,477, standard deviation 34,110, coefficient of variation 16.0%. Quarterly shares: Q1 24.5%, Q2 25.9%, Q3 21.9%, Q4 27.7%, against an even 25.0% each.

Day of week

Day	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Revenue	377,328	360,533	404,038	345,786	299,458	402,405	372,175
Share	14.7%	14.1%	15.8%	13.5%	11.7%	15.7%	14.5%
Transactions	138	146	161	139	123	143	150

Chi-square 5.76 on 6 degrees of freedom, p = 0.45. The weekend carries 30.2% of revenue on 28.6% of the days, which is the share any two days in seven would carry anyway.

Power analysis: how large an export would have to be

This is the one genuinely actionable output of the review, because it converts the bad news into a specification. Simulation was used to ask: if one city genuinely ran a given percentage above the others, how many rows would be needed to detect that reliably, meaning in 80% of samples?

If one city genuinely ran this far ahead	Rows needed for 80% power	Multiple of the current file
50% above the others	1,000	1x, the current size
30% above the others	3,000	3x
20% above the others	6,000	6x
10% above the others	25,000	25x

READ THIS AS A SPECIFICATION

At the current 1,000 rows, the chance of detecting a real 20% lift on one city is **21%**, against the 80% conventionally required. Part of the silence in this file is therefore a sample-size limit rather than proof of a flat business. The honest statement is that the file is too small to settle the question, and the table above says exactly how much larger it would need to be.

8. What may be published, and what to do next

Supported, with the caveats above	Not supported by this file
Total revenue and total units for the year	Any named product or named customer analysis
Monthly revenue trend across 2024	Profit, margin, unit price or basket value
Revenue split by product category, city, gender and age band	Repeat purchase, retention or churn
Transaction counts and average transaction value	Regional or seasonal prioritisation of any kind
	Forecasting, since there is no trend to project

A dashboard restricted to the supported list would be accurate and honest. It would also tell the reader that every category, every city and every month performs the same, which is a description of the file rather than a description of the business. That is the dashboard that was built: six static pages reading this same file, no page ranking a city or a category as best or worst, each city share shown next to its confidence interval rather than as a bare ordering, and the no-signal caveat repeated in every page footer rather than parked on one quality page, because readers arrive at a single page from a link and act on what that page alone appears to say.

Recommendation

- **1. Treat this file as a structural sample, not as sales history.** Its shape, column names and grain are a usable specification for what a real export should contain. Its values are not evidence about the business and should not be quoted as such.
- **2. Request a production export before any decision-support work proceeds.** Four additions would make this dataset decision-grade: a unit cost or unit price column, a stable product key with one fixed category per product, a stable customer record, and at least two years of history so that seasonality can be separated from noise.
- **3. Confirm the intended decision before the build.** The single most useful piece of information we do not yet have is which business decision the dashboard is meant to support. That determines whether the gaps listed above are blocking or tolerable.
- **4. Size the replacement export to the question.** Simulation shows how many rows are needed to detect a genuine difference in one city with 80% confidence: 1,000 rows for a 50% difference, 3,000 for 30%, 6,000 for 20%, and 25,000 for 10%. At the current 1,000 rows the chance of detecting a real 20% difference is 21%. If the business question is which city to prioritise, the export needs to be several times larger than this sample.

Method and evidence

Every figure in this report was computed directly from the delivered file, not estimated, and every one of them is reproducible from the two scripts below.

Evidence	Where it comes from	Reproduce with
Cleaning checklist, key integrity, marginal tests, concentration	Stage 1 to Stage 3 of the assessment	`python 02_Analysis/analyze.py`
Interactions, bootstrap, Gini, Benford, temporal structure, power, anomalies	The seven-angle deep battery	`python 02_Analysis/analyze_deep.py`
This document	Generated from those two outputs, with no figure typed by hand	`python 04_Deliverables/make_report.py`

Statistical tests used: chi-square goodness of fit against a uniform distribution at the 5% level; 20,000-iteration Monte Carlo permutation tests for the city revenue gap and for both revenue interaction matrices; 4,000-resample bootstrap confidence intervals for the city shares; Pearson correlation between revenue and quantity; Gini coefficients on the city, category and transaction distributions; a chi-square leading-digit test against Benford and against a uniform range; lag-1 autocorrelation on the monthly series; and a simulation power analysis at four effect sizes.